Tisha Sehrawat

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EDUCATION

10X International School, Bangalore, India ***Aug 2023-Jun2027***

- International Baccalaureate Diploma Programme candidate (3 sciences)
- **HL:** Biology, Chemistry, Mathematics: Analysis & Approaches | **SL:** Physics, English A Language & Literature, Hindi B

RESEARCH EXPERIENCE

Advanced Program Research Scholar, FARII, mentored by Dr. Robert Austin ***Jun 2024-Present***

Machine Learning Classification of Cancer Types Using 5-Hydroxymethylcytosine (5hmC) Signatures

- Developed a supervised machine-learning framework using promoter-associated 5hmC signatures from cell-free DNA for non-invasive cancer staging; independently built the computational pipeline as the only high-school researcher in the program
- Analyzed 72 variances across 1,000+ patient records with logistic regression, random forest, and gradient boosting models (90-92% accuracy); found a strong negative correlation between 5hmC depletion and cancer progression
- Work independently validated by a Stanford PhD and IISc faculty; collaborated with a Princeton Professor of Biophysics and mentored by a Princeton PhD; attracted seed-stage interest from a NYC venture partner connected to 2A Biosciences

Independent Researcher & Published Author ***Jul 2025-Jul 2026***

Prion Disease in India: A National Review & Diagnostics Analysis

- Authored the first national review of prion disease diagnostics in India, synthesizing 50+ peer-reviewed studies
- Proposed AI-assisted detection, RNA-based therapeutic pathways, and a precision-medicine policy framework; designed a neurologist questionnaire to gather primary data on clinical prion cases
- Published in the Oxford Journal of Student Scholarship, a globally peer-reviewed journal with under 20% acceptance

Research Foundations Scholar, Non-Trivial ***Feb 2026-Apr 2026***

Project TRIAD: The Roots of Intersecting African Diseases

- Selected among <1% of global applicants; trained in hypothesis generation, causal inference, literature synthesis, and research design
- Designed a global health proposal on the convergence of sickle cell anaemia, malaria, and neurodegeneration, linking HbS allele frequency and malaria transmission intensity to long-term neurological outcomes
- Proposed geospatial modeling with IHME and Malaria Atlas Project datasets and a systematic review protocol on malaria outcomes in SCA patients

INNOVATION & ENTREPRENEURSHIP

PlastiXcan, Bangalore, India ***Nov 2025-Present***

Founder, Lead Researcher & Patent Holder

- Built a portable microplastic detection system combining Raspberry Pi imaging, OpenCV computer vision, and optional TensorFlow Lite machine learning at under ₹4,000 per unit
- Traced contamination across 3 Bangalore lakes into food and water; Tested 500+ household samples; reduced testing cost by 95% and improved testing efficiency by 87% versus lab-based FTIR/Raman spectroscopy; endorsed by FSSAI, India's largest food governing body
- Patent published Application No. IN202641095036 A1 (Indian Patent Journal, 14 August 2026); featured in CIO&Leader; won the CREST Gold Award with 100% marks; submitted to the IRIS National Fair 2026

Project CARE: Clinical Anxiety Reduction & Emotional Support ***Jan 2026-Present***

Founder & Developer

- Translated patterns of patient and caregiver anxiety observed during clinical shadowing into an app that reduces fear during diagnosis, surgery, consultations, and waiting periods with multilingual support
- Built around a three-stage framework (Capture Fear, Clarify Uncertainty, Create Comfort) offering emotional check-ins, hospital workflow explanations, and coping tools
- Launched on the Google Play Store in English, Hindi, and Kannada; App Store release upcoming
- Launched in English, Hindi, and Kannada; delivered a TEDx talk, “Fear Is the First Symptom,” with 5K+ views

VisionXcelerate ***Feb 2024-Present***

Co-Founder

- Designed AI-enabled dementia-care glasses with memory prompts, caregiver monitoring, GPS, and fall detection; building an MVP on Mentra glasses
- Synthesized insights from stakeholders (7 physicians, 100+ institutional caregivers, 25+ family caregivers) to ground the product in real-world patient and caregiver needs
- Interviewed 250+ patients and physicians across 3 dementia homes; piloted the system with ~50 patients and achieved 98% fall-detection accuracy
- Validated by neurologists, geriatricians, and AgeTech experts; won the NYAS Wearables Challenge (1st of 167 teams) and reached the Diamond Challenge global semi-finals.

ChoiceX ***Nov 2023-Dec 2025***

Co-Founder

- Built an AI career guidance app for low-income youth in India, using a hybrid recommendation system with TF-IDF vectorization, k-nearest neighbors, and logistic regression to match users with personalized course and job pathways
- Conducted stakeholder research and tested the prototype with 50+ beta users; presented at the EY STEM App Challenge City-Level Showcase and received the Life Entrepreneurship Changemaker Award

CLINICAL EXPERIENCE

Fortis Hospital, under Dr. Ritu Garg ***Jun 2026-Jul 2026***

Clinical Shadow

- Shadowed Neurology, Cardiology, and Oncology across consultations, inpatient care, and specialist discussions as part of a combined 2-week clinical shadowship
- Observed 100+ clinical cases across Narayana and Fortis, including 2+ transplants, 10+ ECMO cases, and 25+ cath-lab procedures

Narayana Health ***Jun 2026-Jul 2026***

Clinical Shadow

- Shadowed across Cardiology, Medical & Surgical Oncology, and Neurology; observed 100+ patient encounters including transplants, ECMO cases, and cath-lab procedures
- Helped build a unified patient registry across three departments, consolidating arrival times, referrals, pending reports, and next destinations; reduced waiting times by 15% and eliminated repeated registration
- Piloted Project CARE with Narayana Health, extending a patient-support tool developed from observed barriers in clinical and community healthcare settings

Aarogya Seva Kendra, NGO Clinic ***Jan 2026-Apr 2026***

Patient Intake & Care Observation

- Assisted with patient intake and observed primary healthcare delivery for underserved populations
- Scaled Project CARE to 4K+ users, with 80% reporting lower anxiety; supported patient intake and vaccination drives for ~200 people

WORK EXPERIENCE

Pharmint Pte Ltd, Singapore ***Jun 2025-Jul 2025***

Bioinformatics & Evolutionary Genomics Intern

- Selected from 500 applicants (~4%); analyzed genotype-phenotype correlations in beak morphology evolution across 30+ avian species using two developmental genes (ALX1, HMGA2)
- Applied multiple sequence alignment, phylogenetic reconstruction, and mutation mapping; processed 5,000+ genetic sequences and built 1,000+ evolutionary trees with ~95% accuracy in identifying conserved regions
- Identified 8+ significant mutations and 3 cases of convergent evolution; improved sequence-curation reliability by 75%

SankalpTaru Foundation (CROASIS) ***Jun 2025-Sept 2025***

Project Intern

- Prototyped a Carbon Credit Tracking Dashboard in Figma for credit estimation and emissions monitoring; authored the user guide and contributed vendor and cost research to the roadmap
- Evaluated vendor options and cost estimates to identify potential savings of 15–20% and improve budget efficiency for dashboard implementation.

LEADERSHIP & COMMUNITY SERVICE

Student Council, 10X International School ***Oct 2025-Present***

House Captain

- Led a 40+ member house team and organized 5+ academic, cultural, and athletic events; mentored house members and supported school-wide initiatives
- Supported 120+ students through the Innovation Council, contributing to STEM initiatives, student innovation, and school-wide programming

The Scientific Trail, 10X International School ***May 2026-Aug 2026***

Founder & Editor-in-Chief

- Founded the school's first student-led science magazine, engaging 60+ students in writing on biomedical research and emerging technologies
- Built an editorial platform for student research and science communication, supporting scientific writing, research exploration, and publication across the school community

Science Club 'The Significant Figures', 10X International School ***Aug 2025-May 2026***

Co-Founder & Co-President

- Co-founded and lead a 30+ member platform promoting scientific inquiry; organize workshops, discussions, and peer-led learning initiatives

- Guided 50+ students through STEM workshops, research writing, and science communication while supporting collaborative learning across the school

Shiksham: Empowering Rural Girls Through Science, Mentorship & Holistic Learning ***Oct 2023-May 2025***

Co-Founder

- Curated STEM curriculum for 70+ girls across 10 villages through workshops and mentorship; expanded reach through NGO partnerships
- Secured and channelled ₹10,000+ toward books, stationery, and academic resources; managed tutoring programs and mental-wellness support; Led 12+ mentors in tutoring and mental-wellness support

10X International School ***Oct 2024-Present***

Additional Roles

- **Class Head, Grades 9 & 10** (25+ students) | **Innovation Council Member & Hercules Innovation Captain**
- **Editorial Team, 10Xpress** and **Editor, Yearbook Committee** (2024-2025): wrote health and medicine pieces; led yearbook editorial planning
- **Member, International Youth Neuroscience Association** (2025-2026): contributed to global neuroscience outreach

SUMMER & PRE-COLLEGE PROGRAMS

Columbia University Irving Medical Center, New York, NY ***June 2026-July 2026***

Scientific Enrichment Month (SEM) Research Scholar

- Selected for an enrichment program in cancer research and oncology; attended expert-led sessions on cancer biology, translational and precision medicine

Harvard Student Agencies ***May 2024-July 2024***

Medical Research Scholar, Future Doctors I

- Studied genetics, immunology, endocrinology, and organic chemistry; analyzed patient data in clinical simulations for differential diagnosis

Science Mentorship Institute ***July 2024-August 2024***

Scientific Researcher, Biology Mentorship Program

- Conducted an independent research project and presented papers in weekly journal clubs

ICREST Academy ***April 2024-July 2024***

Summer Researcher, Advanced Learner's Program (ALP 101 & 201)

- Completed hands-on programs in microbiology, biochemistry, immunology, molecular biology, and bioinformatics

SKILLS & TRAINING

- **Programming:** Python, R, HTML, CSS, JavaScript
- **Software & Hardware:** Figma, OpenCV, TensorFlow Lite, MATLAB, Raspberry Pi
- **Research & Communication:** research design, literature synthesis, scientific writing, stakeholder analysis, public speaking
- **Courses:** Johns Hopkins Cancer Biology (2025); Duke University Medical Neuroscience (2025); UChicago Understanding the Brain (2025); Leaf Program, Biology for a Better Tomorrow (2026); Introduction to Programming Using Python, Udemy (2024); Full Innovation Course, StartupYou (2023); Young Member, NYAS (2024)
- **Languages:** English (full professional), Hindi (native/bilingual), Korean (elementary)

AWARDS

NYAS Junior Academy Wearables Challenge: Global Winner, 1st of 167 teams (2024)
CREST Gold Award, British Science Association: Gold, 100% marks for PlastiXcan (2026)
International Biology Bowl: Bronze Medal, 3rd Place (2026)
International Chemistry Competition: Finalist (2026)
Pulitzer Center Local Letters for Global Change: Global Finalist, Global Health, from ~15,000 entries (2026)
Diamond Challenge: Global Semi-Finalist, Team VisionXcelerate (2026)
Yale International Relations Association: Honorable Mention (2025)
InnovateXHarrow Business Pitch Challenge: Runner-Up, BioBeats (2024)
National Astronomy Challenge: Gold Medal, National Rank 32 (2023)
Stonehill Science Bowl: Runner-Up, Modern Problems Category (2023)

ATHLETICS & CREATIVE WORK

- **Athletics:** state-level swimmer representing Telangana; gold medalist in football; active badminton player
- **Music & Dance:** guitarist, vocalist, and composer of original music; award-winning solo and group dance performer